Nagar Yuwak Shikshan Sanstha's
Yeshwantrao Chavan College of Engineering
(An Autonomous Institution affiliated to Rashtrasant Tukadoji Maharaj Nagpur University)
Hingna Road, Wanadongri, Nagpur - 441 110
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Ph.: 07104-237919, 234623, 329249, 329250 Fax: 07104-232376, Website: www.ycce.edu

Department of Computer Technology

**Vision Statement of the Department.**

**To be a preferred hub for nurturing computer engineering aspirants through innovative, ethical, value-based technical education, impactful research, and industry collaboration, to become successful professionals contributing for the betterment of society.**

**Mission Statement of the Department.**

**To nurture computer engineering professionals by offering Dynamic learning environment that encourages problem-solving using emerging technology, National Education Policy aligned evidence-based technical education, fostering ,innovation, research, and industry–institute collaboration & Holistic, ethical and sustainable development for lifelong success.**

**Session 2026-2027**

| **Vision:**Dream of where you want. | **Mission:** Means to achieve Vision |
|---|---|

**Program Educational Objectives of the program (PEO):** (broad statements that describe the professional and career accomplishments)

| | | | |
|---|---|---|---|
| PEO1 | **Preparation** | **P: Preparation** | **Pep-CL abbreviation pronounce as Pep-si-lL easy to recall** |
| PEO2 | **Core Competence** | **E: Environment (Learning Environment)** | |
| PEO3 | **Breadth** | **P: Professionalism** | |
| PEO4 | **Professionalism** | **C: Core Competence** | |
| PEO5 | **Learning Environment** | **L: Breadth (Learning in diverse areas)** | |

**Program Outcomes (PO):** (statements that describe what a student should be able to do and know by the end of a program)

**Keywords of POs:**
Engineering knowledge,Problem analysis, Design/development of solutions,Conduct Investigations of Complex Problems, Engineering Tool Usage, The Engineer and The World,Ethics, Individual and Collaborative Team work,Communication, Project Management and Finance, Life-Long Learning

**PSO Keywords:** Cutting edge technologies, Research

"I am an engineer, and I know how to apply engineering knowledge to investigate, analyse and design solutions to complex problems using tools for entire world following all ethics in a collaborative way with proper management skills throughout my life." *to contribute to the development of cutting-edge technologies and Research*.

**Integrity:** I will adhere to the Laboratory Code of Conduct and ethics in its entirety.



07/07/2026

**Name and Signature of Student and Date**
(Signature and Date in Handwritten)

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| Session | 2025-26 (ODD) | Course Name | Web Technology Lab |
|---|---|---|---|
| Semester | 3 | Course Code | 23CT1301 |
| Roll No | 22 | Name of Student | Sanskruti . P . Kharapkar |

| Practical Number | 8(B) |
|---|---|
| Course Outcome | 1. Understand various internet technologies.<br>2. Design the web pages using HTML and CSS.<br>3. Implement the XML technology to store the data.<br>4. Develop the interactive web pages using JavaScript. |
| Aim | To develop a Typed To-Do List application using React and TypeScript with reusable components, props, and the `useState` hook. |
| Problem Definition | To develop an interactive To-Do List application that allows users to add tasks, mark them as completed, and delete them. The application uses a common Task interface to maintain the structure of task objects. React components, props, and state management are used to update the task list dynamically without reloading the webpage. |
| Theory<br>(100 words) | React is a JavaScript library used to build interactive user interfaces through reusable components. TypeScript adds static typing to JavaScript and helps identify type-related errors during development. In this practical, the Task interface defines the structure of each task using an ID, title, and completion status. Props pass data and callback functions from parent components to child components. The `useState` hook stores the task list and updates the interface when the state changes. The `map()` method displays tasks, while `filter()` removes selected tasks. Event handlers connect buttons to |

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| | |
|---|---|
| | **task operations, making the application interactive and easier to maintain.** |
| Procedure and Execution<br><br>(100 Words) | Step for Implementation:<br><br>1. **Create a React + TypeScript project using Vite.**<br>2. **Define the Task interface in `types.ts`.**<br>3. **Create `TaskInput.tsx` and `TaskItem.tsx` components.**<br>4. **Implement task management and state handling in `App.tsx`.**<br>5. **Connect the application through `main.tsx`.**<br>6. **Design the interface using `App.css` and `index.css`.**<br>7. **Run the application using `npm run dev`.**<br>8. **Test adding, completing, and deleting tasks.** |
| | Code:<br><br>• **`index.html` — provides the HTML entry point for React.**<br><br>`<!doctype html>`<br>`<html lang="en">`<br>`<head>`<br>`<meta charset="UTF-8" />`<br>`<meta name="viewport" content="width=device-width, initial-scale=1.0" />`<br>`<title>Typed To-Do List</title>`<br>`</head>`<br>`<body>`<br>`<div id="root"></div>`<br>`<script type="module" src="/src/main.tsx"></script>` |

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```
  </body>

</html>
```

- **`src/types.ts` — defines the shared Task interface.**

```
export interface Task {

    id: number;

    title: string;

    completed: boolean;

}
```

- **`src/components/TaskInput.tsx` — accepts task input and provides the Add Task button.**

```
import { useState } from "react";



interface TaskInputProps {

    onAdd: (title: string) => void;

}



function TaskInput({ onAdd }: TaskInputProps) {

    const [title, setTitle] = useState<string>("");



    function handleAdd(): void {
```

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```
        if (title.trim() === "") {
            return;
        }

        onAdd(title.trim());
        setTitle("");
    }

    return (
        <div>
            <input
                type="text"
                value={title}
                onChange={(event) =>
                    setTitle(event.target.value)
                }
                placeholder="Enter task"
            />

            <button onClick={handleAdd}>
                Add Task
```

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```
            </button>

        </div>

    );

}



export default TaskInput;
```

- **`src/components/TaskItem.tsx` — displays each task with Complete and Delete buttons.**

```
import type { Task } from "../types";



interface TaskItemProps {

    task: Task;

    onToggle: (id: number) => void;

    onDelete: (id: number) => void;

}



function TaskItem({

    task,

    onToggle,

    onDelete
```

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```
}: TaskItemProps) {

    return (

        <li>

            <span

                style={{

                    textDecoration: task.completed

                        ? "line-through"

                        : "none"

                }}

            >

                {task.title}

            </span>


            <button onClick={() => onToggle(task.id)}>

                Complete

            </button>


            <button onClick={() => onDelete(task.id)}>

                Delete

            </button>

        </li>
```

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```
    );

}



export default TaskItem;
```

- **`src/App.tsx` — manages task data using `useState` and renders task components.**

```
import { useState } from "react";

import TaskInput from "./components/TaskInput";

import TaskItem from "./components/TaskItem";

import type { Task } from "./types";

import "./App.css";



function App() {

    const [tasks, setTasks] = useState<Task[]>([]);



    function addTask(title: string): void {

        const newTask: Task = {

            id: Date.now(),

            title: title,

            completed: false

        };
```

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```
    setTasks((previousTasks) => [
        ...previousTasks,
        newTask
    ]);
}

function toggleTask(id: number): void {
    setTasks((previousTasks) =>
        previousTasks.map((task) =>
            task.id === id
                ? {
                    ...task,
                    completed: !task.completed
                }
                : task
        )
    );
}

function deleteTask(id: number): void {
```

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```
        setTasks((previousTasks) =>
            previousTasks.filter(
                (task) => task.id !== id
            )
        );
    }

    return (
        <div className="todo-app">
            <h1>Typed To-Do List</h1>

            <TaskInput onAdd={addTask} />

            {tasks.length === 0 ? (
                <p>No tasks yet. Add your first task!</p>
            ) : (
                <ul>
                    {tasks.map((task) => (
                        <TaskItem
                            key={task.id}
                            task={task}
```

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```
                                onToggle={toggleTask}
                                onDelete={deleteTask}
                            />
                        ))}
                    </ul>
                )}

                <p>Total tasks: {tasks.length}</p>
            </div>
        );
}

export default App;
```

- **`src/main.tsx` — renders the main React component.**

```
import { StrictMode } from "react";
import { createRoot } from "react-dom/client";
import "./index.css";
import App from "./App";

createRoot(document.getElementById("root")!).render(
```

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```
    <StrictMode>

        <App />

    </StrictMode>

);
```

- **`src/App.css` and `src/index.css` — style the application.**

```
.todo-app {
    max-width: 600px;
    margin: 40px auto;
    padding: 30px;
    background: white;
    border-radius: 12px;
    box-shadow: 0 4px 15px #dce3ec;
}

.todo-app h1 {
    color: #17365d;
    margin-bottom: 25px;
}

.todo-app input {
    width: 65%;
    padding: 11px;
    border: 1px solid #cbd5e1;
    border-radius: 6px;
    font-size: 15px;
}

.todo-app button {
    padding: 10px 13px;
    margin-left: 8px;
    border: none;
    border-radius: 6px;
```

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```
    background: #17365d;
    color: white;
    cursor: pointer;
}

.todo-app button:hover {
    opacity: 0.85;
}

.todo-app ul {
    list-style: none;
    padding: 0;
    margin-top: 25px;
}

.todo-app li {
    display: flex;
    align-items: center;
    gap: 8px;
    margin: 12px 0;
    padding: 12px;
    background: #f1f5f9;
    border-radius: 7px;
}

.todo-app li span {
    flex: 1;
    overflow-wrap: anywhere;
}

.todo-app li button {
    margin-left: 0;
    padding: 7px 10px;
}

.todo-app li button + button {
    background: #b83232;
```

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```
}

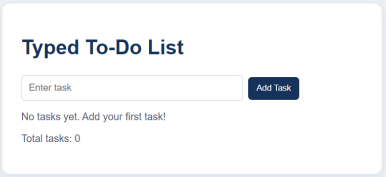
.todo-app p {
    color: #536176;
}

@media (max-width: 500px) {
    .todo-app {
        margin: 20px 12px;
        padding: 18px;
    }

    .todo-app input {
        width: 55%;
    }

    .todo-app li {
        flex-wrap: wrap;
    }
}
```

Output:

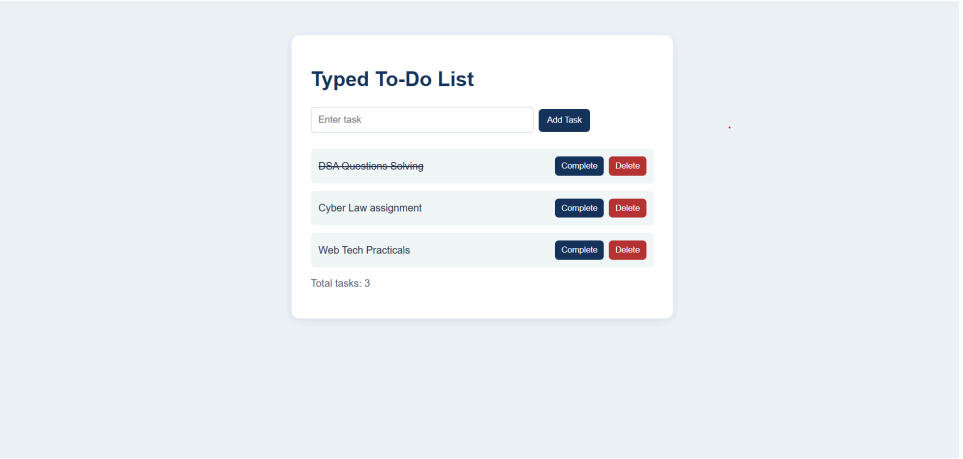


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| |  |
| | |
| Output Analysis | **The application demonstrates React components, props, TypeScript interfaces, and state management using `useState`. The Task interface ensures a consistent structure for task objects. The `map()` method renders task components, while `filter()` removes selected tasks. Event handlers allow users to perform operations through buttons. The interface updates dynamically when the state changes, demonstrating the integration of React and TypeScript in an interactive application.** |
| Link of student Github profile where lab assignment has been uploaded | **https://github.com/Sanskruti2007-code/web-Technology-Practicals** |

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| Conclusion | **The Typed To-Do List application demonstrates how React and TypeScript can be used to build an interactive task management interface. It allows users to add, complete, and delete tasks dynamically. This practical provides experience with reusable components, props, interfaces, the `useState` hook, and array methods. It also demonstrates how type safety and state management help create structured and maintainable web applications.** |
| Plag Report (Similarity index < 12%) | **Similarity index 10%** |
| Date | **06/10/26** |